

LLM 2-Days Report

Chat GPT ni 1 marta train qilish uchun 100 mln\$ dan ko'p ketadi

LM - language Model.

LLM - esa large language model.

Emergence - training boray up, daroni itgach birdaniga natija yaxshilanib boshlash xalati.

RLHF - reinforcement learning human feedback.

Words vs Token

Token - bu berilgan gapni LLM tokenlarga solib olishi.

1 ta token = 0.75 words ga teng
bu degani 100 ta token 75 ta
so'zga teng bo'ladi. O'qiladigan
token va words soni teng
bo'lishi ham mumkin.

3 xil holat mavjud

1 = Token = Words.

2 = Token > Words.

3 = Token < Words.

Token < Words bo'lsa, turdala
barg qiladi =>

English = 1 Token $\approx$ 0.75 words

Spanish / French = 1 Token = 0.80 words

Russian = 1 Token = 0.88 words

Japanese / Korean / Hindi = 1 Token = 0.78 words

LLM lar quyidagi foydalar uchun Tokenlar foydalanadi

- Xotirani tejaydi
- yangi va ilgari ko'rilgan so'zlarni belgilab tushiradi
- turli tillar b-n samarali ishlays
- kod, belgi, va aralash ma'nini bir xil usulda qayta ishlash
- Transformer arxiv tuzilishi uchun ragamli token ID va embeddinglar bilan ishlash ancha qulay.

Tensor - Continuous memoryda saqlanadigan Deep RNNlar sonlar massivi. DL dagi hamma narsani ushbu strukturali tensor bilan

LLM ni train qilishdan oldin quyidagilarni xisoblashimiz kerak

- 1- Parametrlar
- 2- Xotiraga saqlashni
- 3- Compute budjeti
- 4- Layer (wall-clock)
- 5- Nvtx
- 6- Samaradorlik

Bining model namunosa

Layers (L) 24

Model (M) - 2018

Kontekst usunlogi 2018 token

Precision = BF16 compute. FP32 state

Optimizer - AdamW

Hardware = 4x A100 (80 GB)

Demak bo'lsa 1.21 mlrd parametrs

Xotira boyicha 16B + 1.21B $\approx$ 13.3B

Demak xotiraga sifqada

Xotiraga sifqada parametrs/weight

Weight xisobida taxminan 87 saqlan

Weight ketak ekan. (4 kun)

Max boyicha saqlash = 6008 va 8008

pul ketadi ekan. 807 marta

train uchun.

Samaradorlik rishaglar

Bo'z sol hardware samaradorlik

① MFU ni sharoit

② BF16 (FP16 emas)

③ Activation checkpoint

④ Batch / shape tanlash

CPU - u y osh xonasi (moslashuv ekan, sekun)

GPU - samarad savodi Datani

CPU da yuklang, for marta GPU ga
ko'chirib, u yerda saqlang.

Contiguous Tensor - elementlari
memoriyada bitta uzluksiz blokda
row-major tartibda saqlanadigan
tensor

Maximal GPU - throughput ning
deyarli cho'qqisiga chekkadi.

MFU - model flops utilization
GPU dan xarakteristik samovoli for
dolandaligot, ingizni olchaydi.

LLM creation process

Begin project → Project Hardware Setup →
→ Data Engineering → Model
Architecture & Tokenization →
→ Pre-training Configuration →
Execution → Fine tuning & Alignment
→ Deployment.